



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Simplify Algebraic Expressions

Lesson 6-15 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can simplify algebraic expressions by combining like terms.

Language Objective: I can explain my steps using the words like terms, combine, simplify, and coefficient.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Simplify Algebraic Expressions

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property



Tap any word to see what it means and a picture.

1

Combine to write a shorter expression with the same value.

2

Terms with the same variable part, like $3y$ and $8y$, are

.

3

To add or subtract like terms into a single term is to

 them.

4

Rewriting an expression in a shorter form that is still equal to it is to

 it.

5

The number multiplied by a variable is the .

6 A number or variable separated by + or – signs is a .

7 Multiplying each part of a sum by a number uses the property.

Write About the Math

Simplify the total order expression. What does each part of the simplified expression tell you?

Simplifica la expresión del pedido total. ¿Qué te dice cada parte de la expresión simplificada?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Simplify Algebraic Expressions**

Simplificar expresiones algebraicas

☐ **Like Terms**

Términos semejantes

☐ **Combine**

Combinar

☐ **Simplify**

Simplificar

☐ **Coefficient**

Coeficiente

Model: You can only combine items of the same type, just like like terms. — Students explain microphones combine with microphones ($4 + 2 + 6 = 12$) but not with stands, connecting this to combining like terms in algebra.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The total order simplifies to** ____ **m +** ____ **. The** ____ **tells us** ____ **and the** ____ **tells us** ____.
- **My answer is** ____ **because** ____.

Mi respuesta es ____ *porque* ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** ____.
- **My evidence is** ____, **so** ____.

Mi afirmación es ____.

Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Simplify Algebraic Expressions
2. **Notes 2:** Like Terms
3. **Notes 3:** Combine
4. **Notes 4:** Simplify
5. **Notes 5:** Coefficient
6. **Notes 6:** Term
7. **Notes 7:** Distributive property
8. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms — for example, simplifying $6x + 4 + 2x$ as $12x$ by adding all three numbers ($6 + 4 + 2 = 12$) instead of recognizing that 4 has no variable and cannot combine with $6x$ and $2x$. The correct simplified expression is $8x + 4$ (combine $6x + 2x = 8x$, and let the 4 stay on its own). Before you submit, check that every term you combined has the exact same variable part.*
9. **Try It 1:** A. $4y$ — Subtract the coefficients of the like terms: $7 - 3 = 4$. Keep the variable y , giving $4y$.
10. **Try It 2:** A. $9m + 4 - 6m + 3m = 9m$. The constant 4 has no like term, so it stays. Result: $9m + 4$.